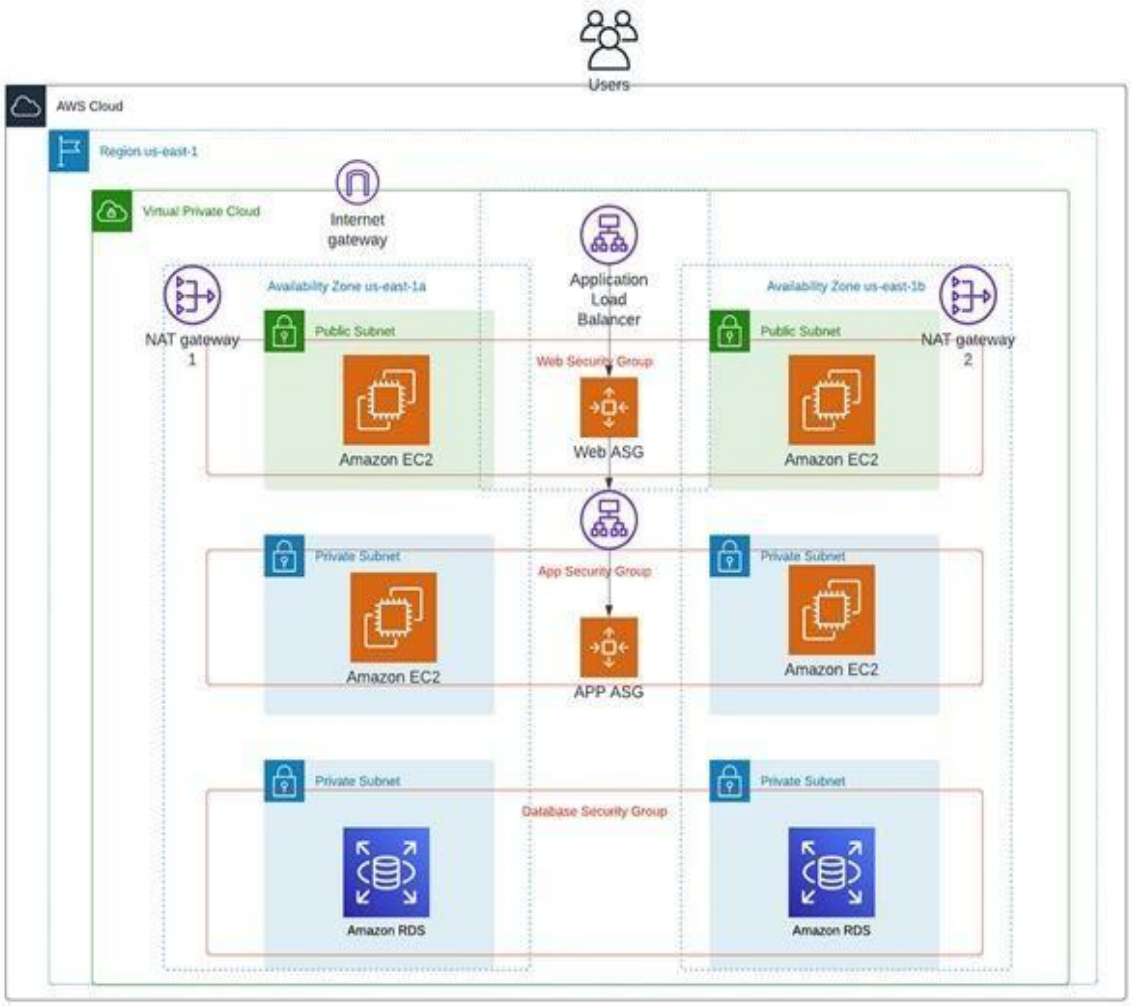
AWS PROJECT

Lavanya.G

Three tier architecture project



Three tier-Architecture diagram

The above picture shows the Three-Tier-Architecture for AWS ➢ First tier is a web tier. It consists of 2 public subnets in different availability zones, and an Auto scaling group with launch template and security group.

➢ The second tier is an application tier. This tier will consist of 2 Private subnets, an Autoscaling group with launch a template and the security group is same.

➢ The third tier is a Database tier. This tier will have an RDS (relational database service) in 2 Private subnets and has the same security group.

security group port numbers:

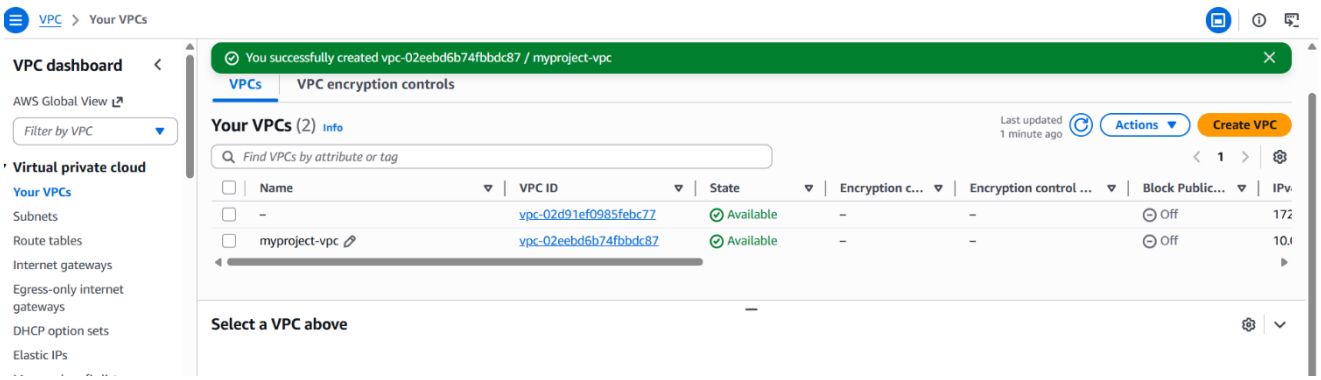
1.SSH = 22

2.HTTP=80

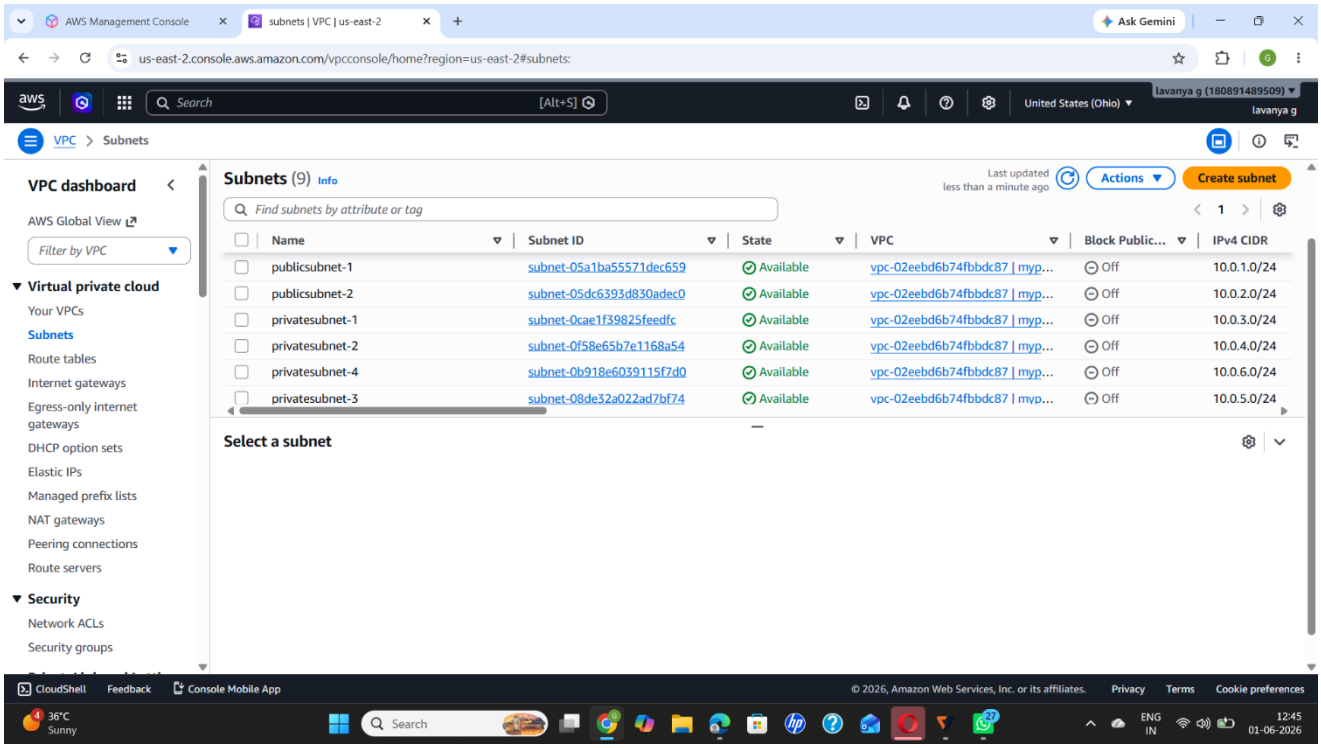
3.MYSQL/AURORA=3306.

STEP 1:

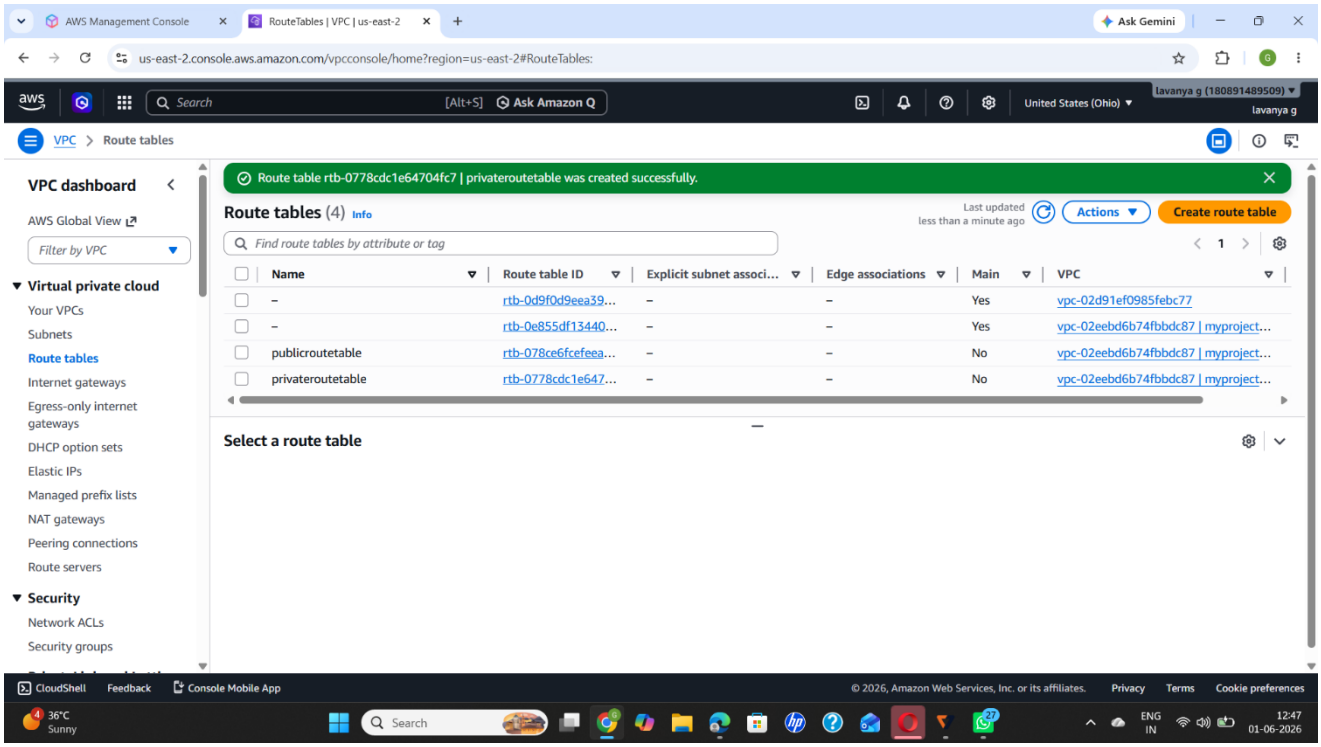
. create a vpc (myproject-vpc)



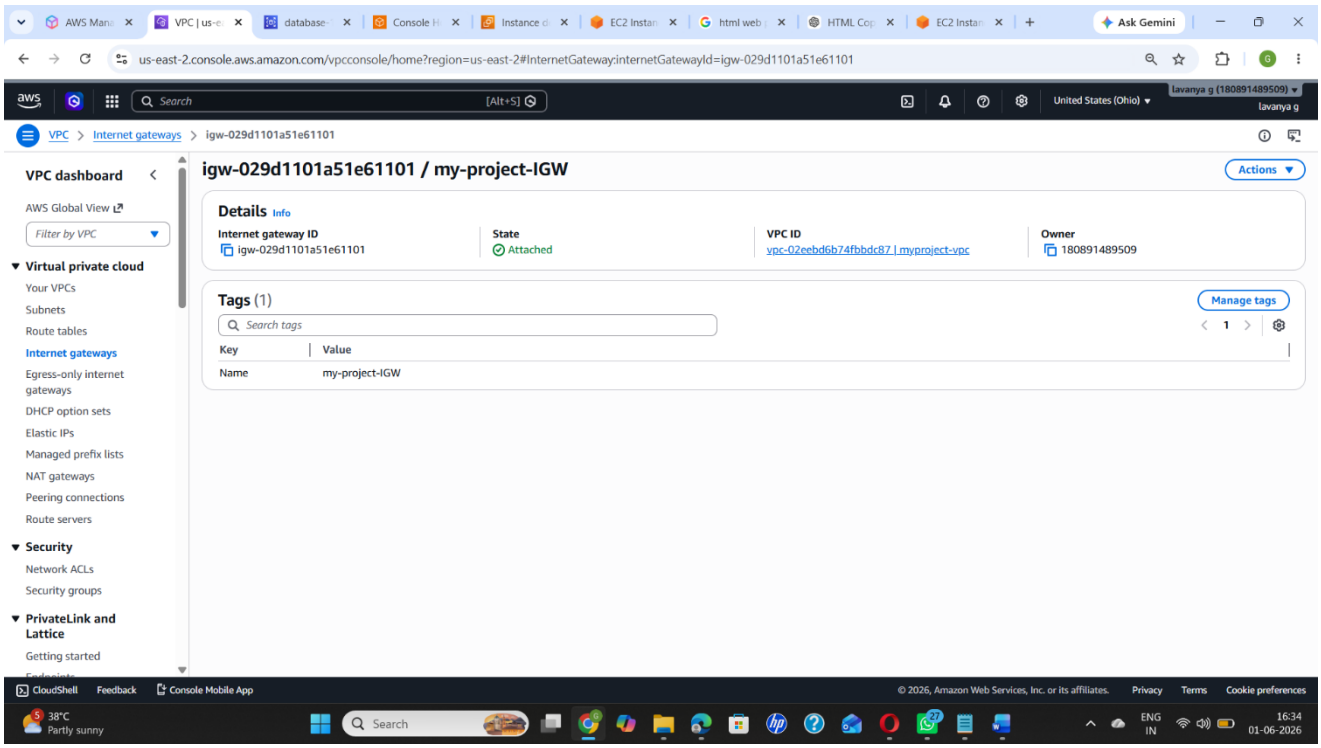
. create 6 subnets in which 2 are public subnets,4 private subnets with different availability zones.



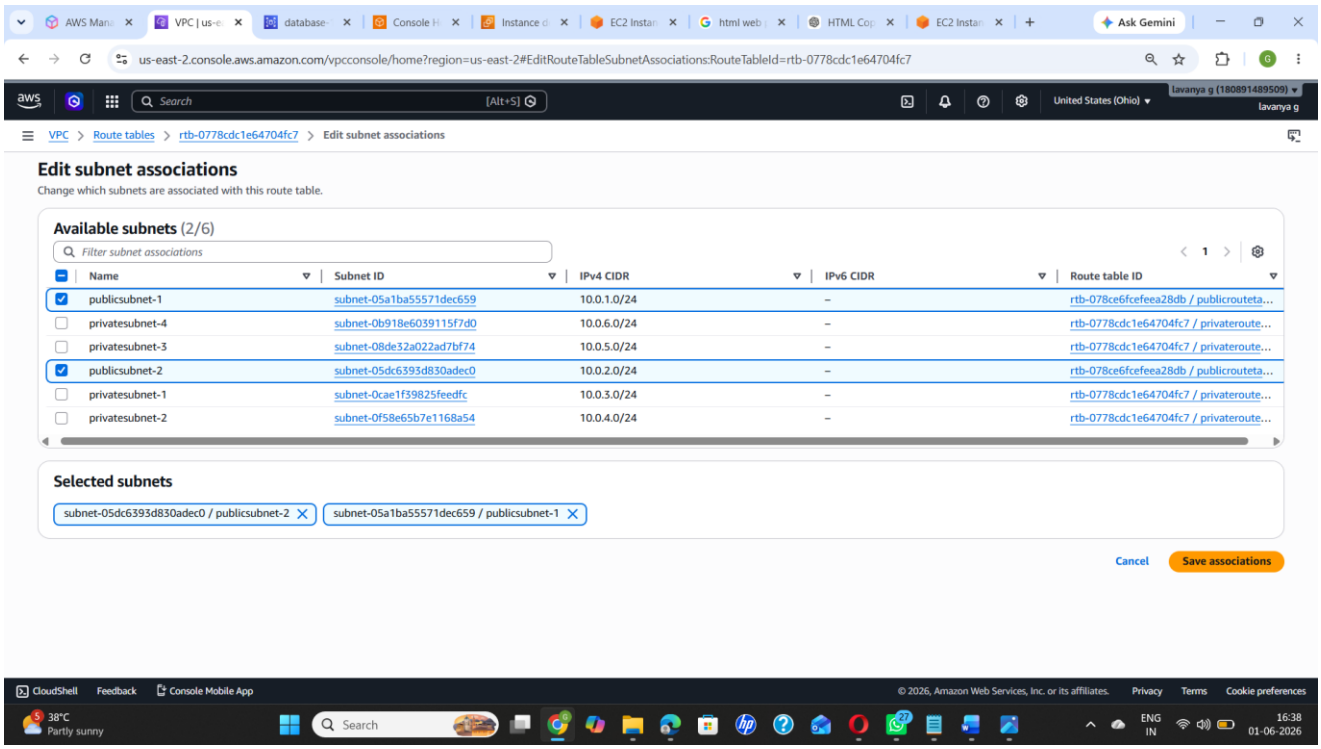
. create two route tables one public route table,one private route table.

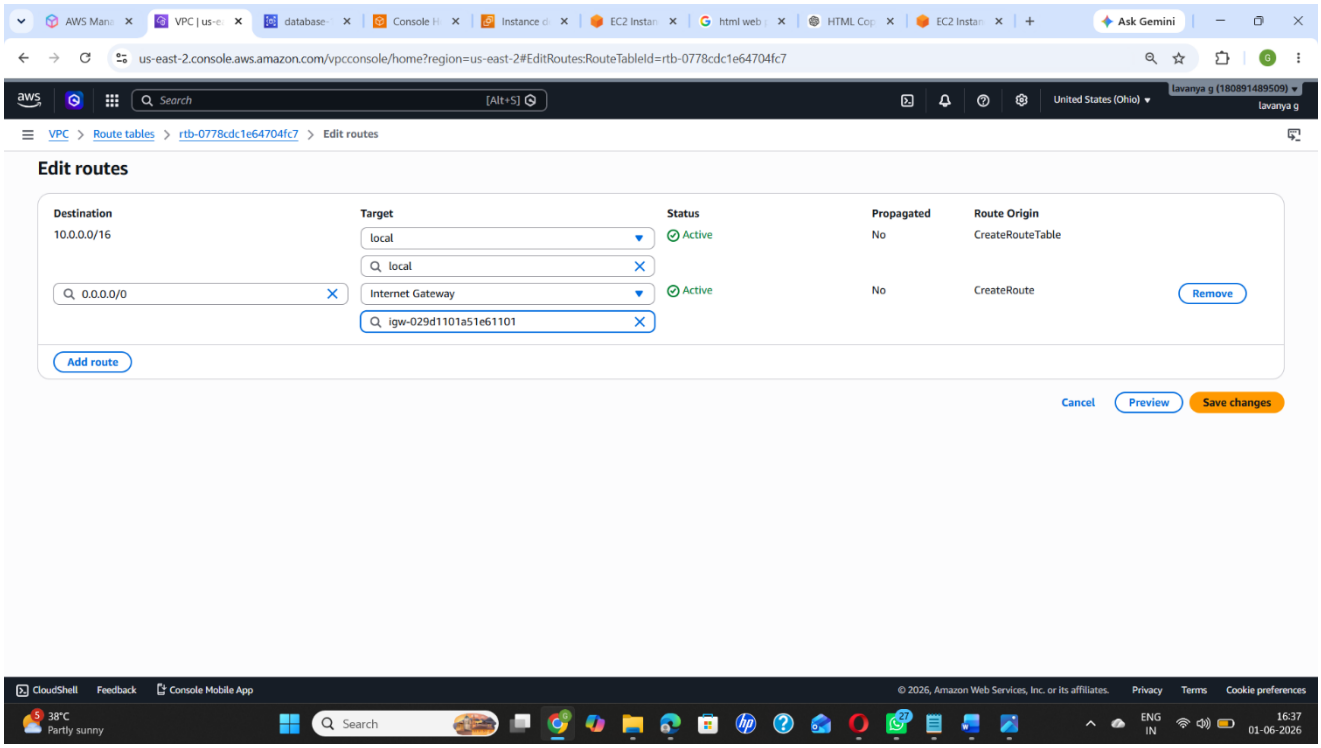


. now create an Internet Gateway and attach it to the created VPC.

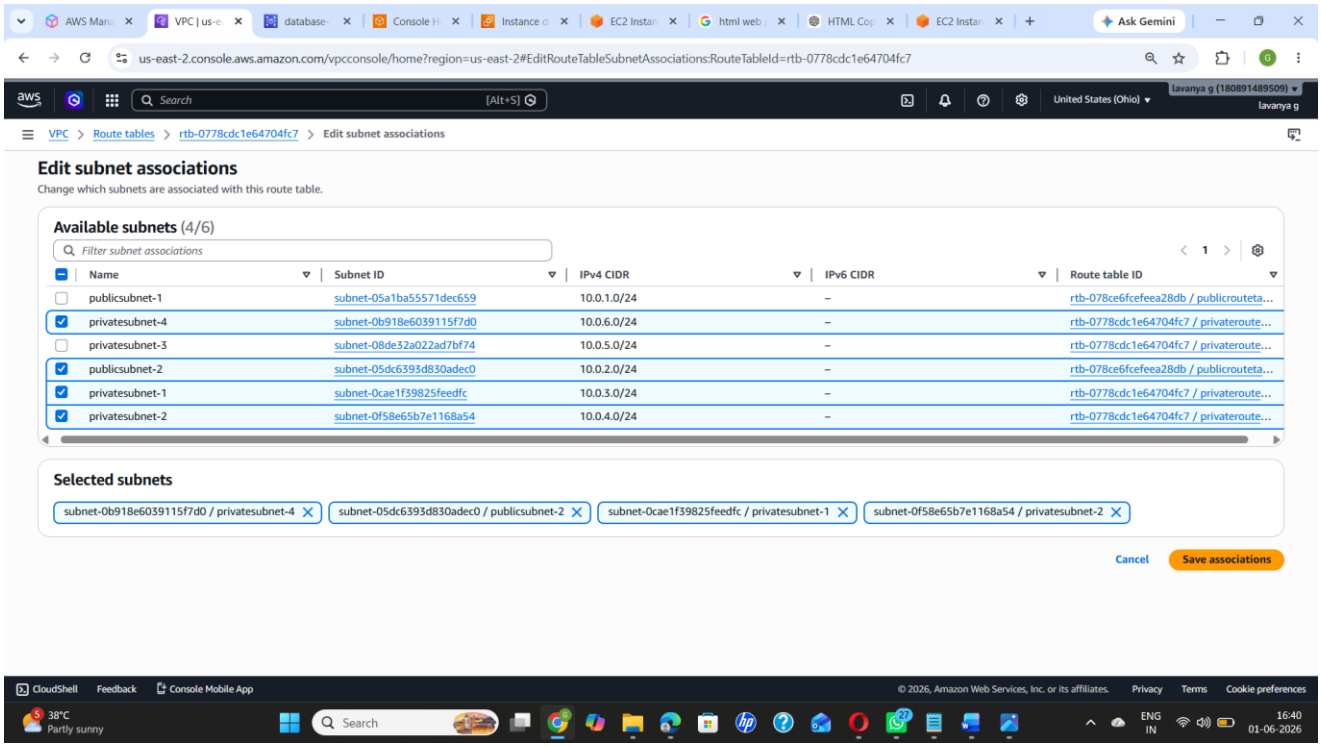


. Now go to subnet associations in public route table click on edit then add 2 public subnets to it. Go to edit routes and Attach internet gateway to it.

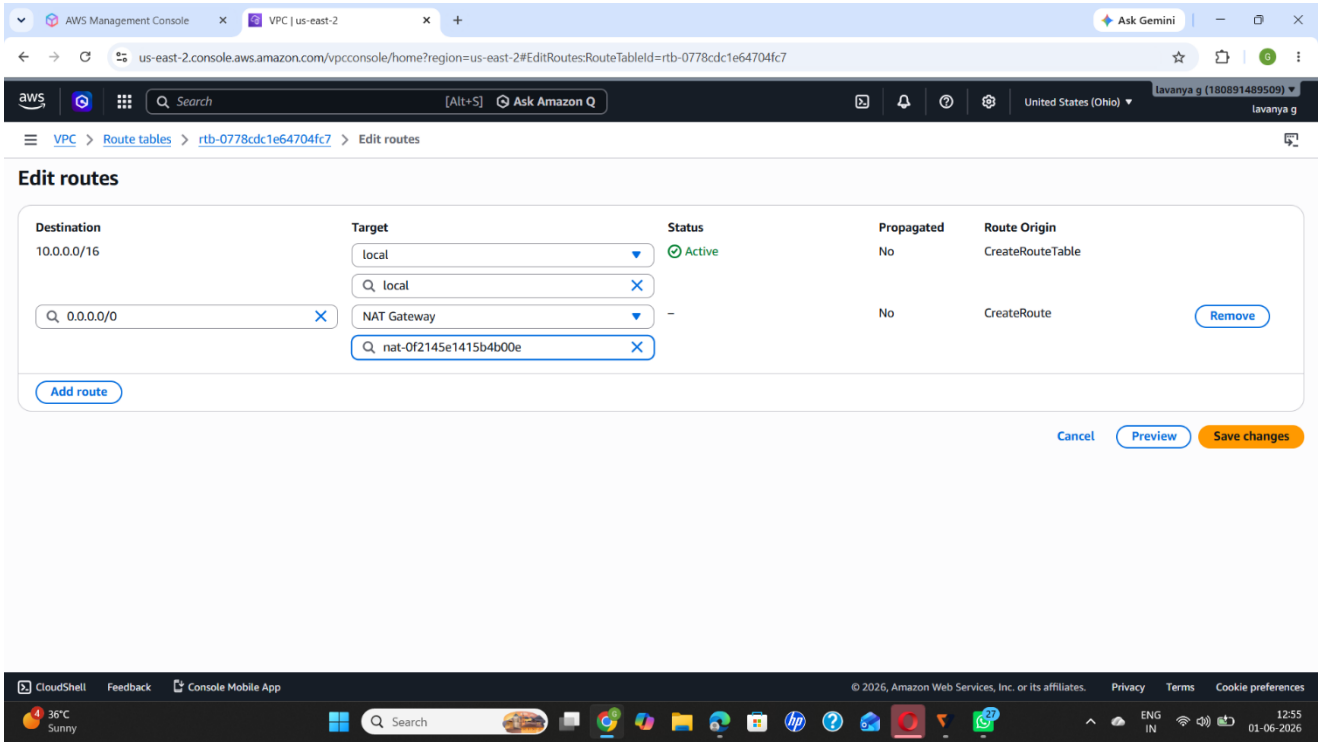




. Now go to subnet associations in private route table add 4 private subnets to it.



. Now create NAT Gate Way using public subnet, then attach it to the private route table. Also allocate elastic IP to it.



Step 2:

Now create 4 ec2 instances.

2 public instances,2 private instances with different avalability zones.

1.give name (publicinstance-1)

2.select Amazonlinux/ubuntu as your wish.

3.select keypair

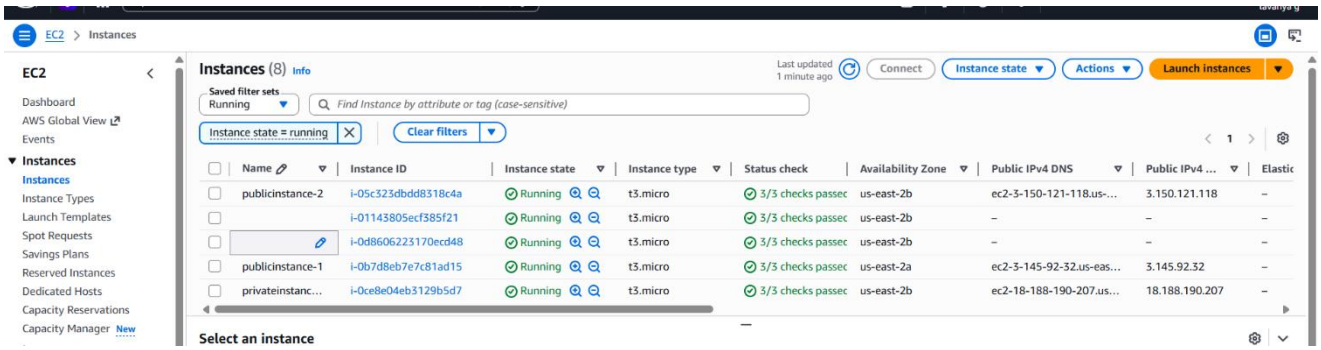
4.select created vpc(myproject-vpc)

5.select avalability zone

6.enable auto assign public IP

7.create a security group(add above mentioned ports)

8.click on launch instance.



Now login into public-instance 1 in my case I have taken Ubuntu.so I need to install Apache2.

1.apt update -y.

2.apt install apache2 -y.

3. systemctl status apache2.

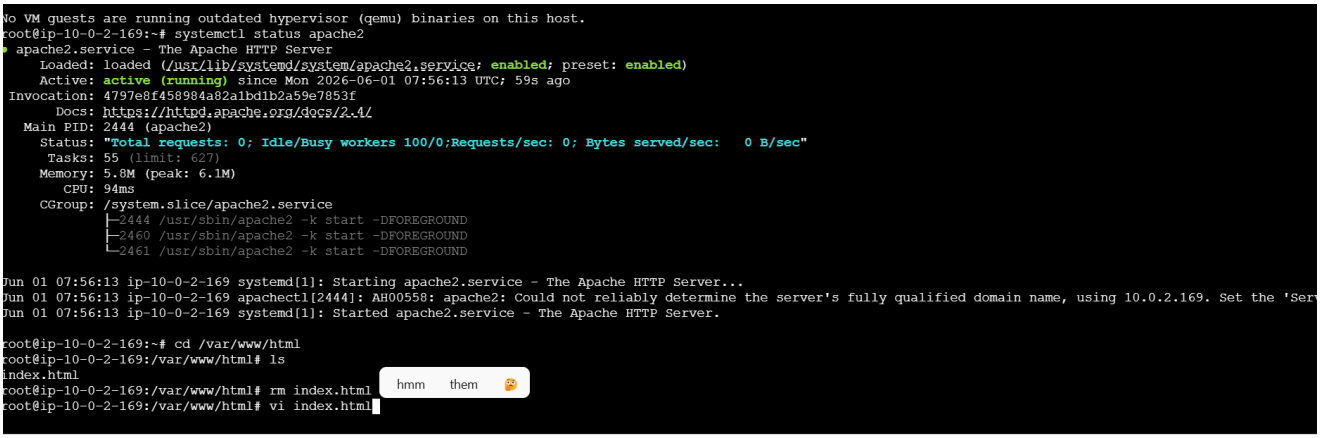
4. cd /var/www/html.

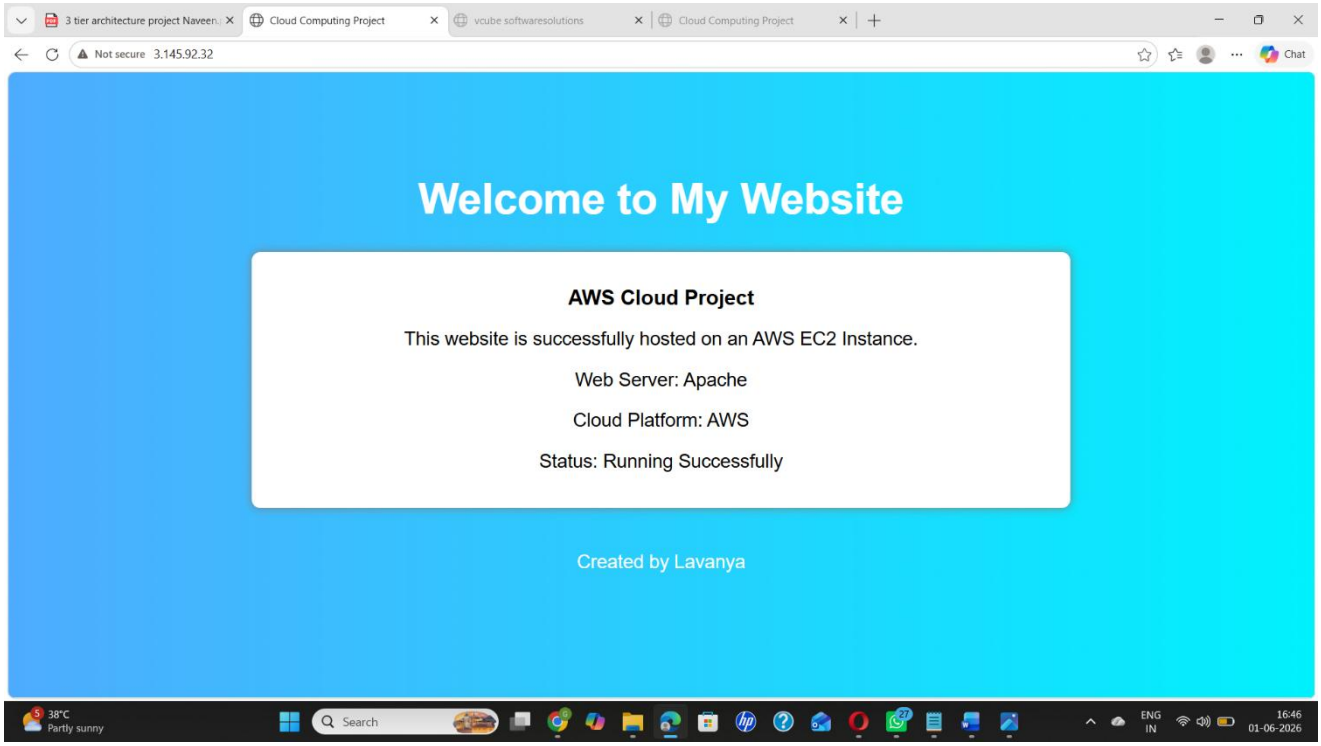
5. rm index.html

6. vi index.html

7.systemctl restart apache2.

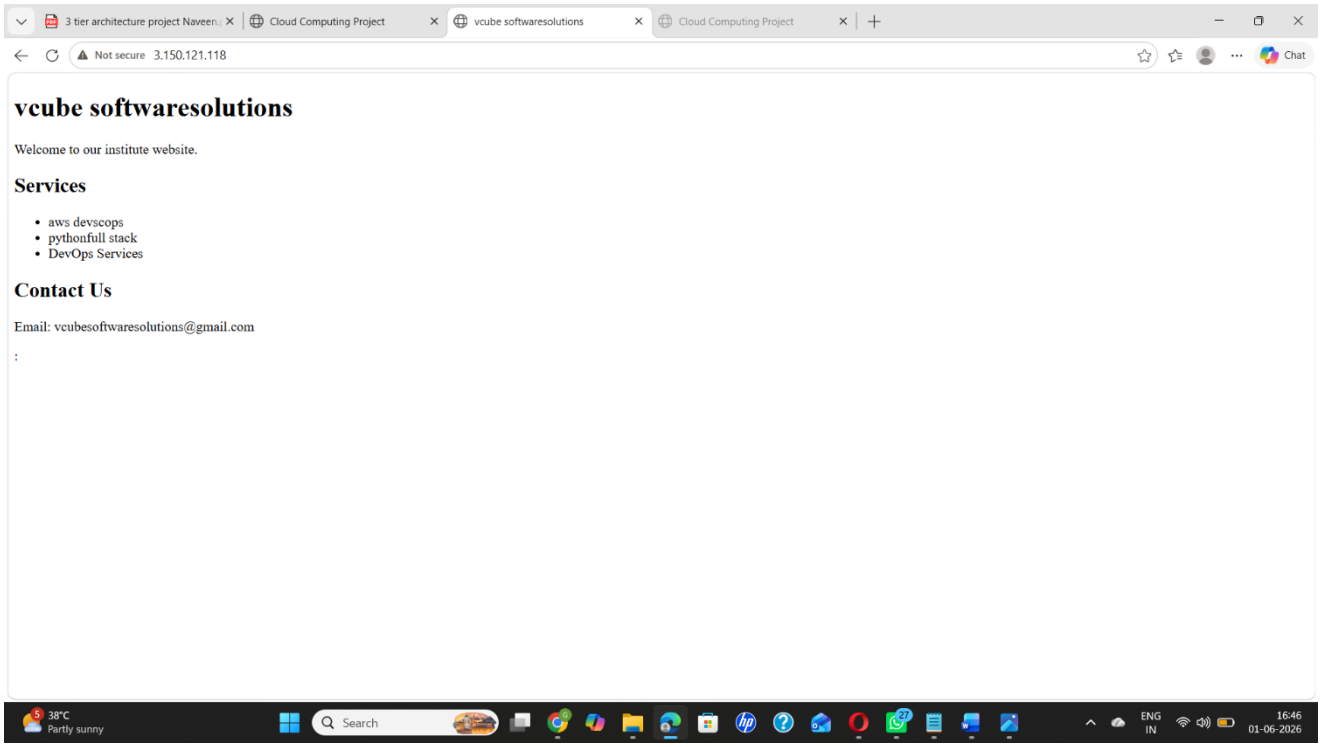
Now copy the public ip of instance and paste it on the new tab it will show the information which u have entered.





now do the same with public-instance 2.

Because I have taken ubuntu in all instances.

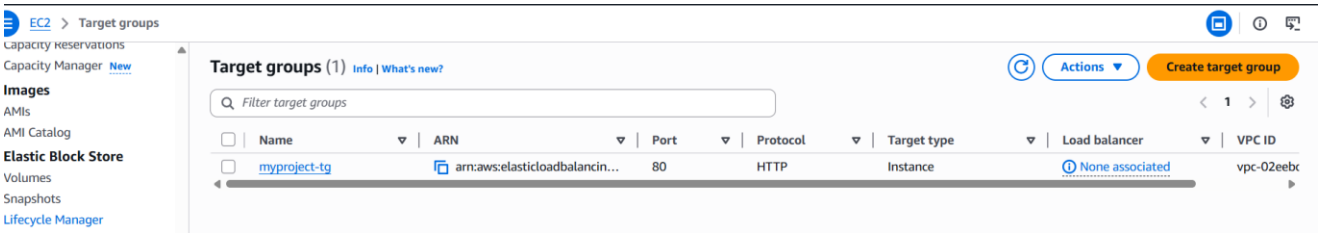


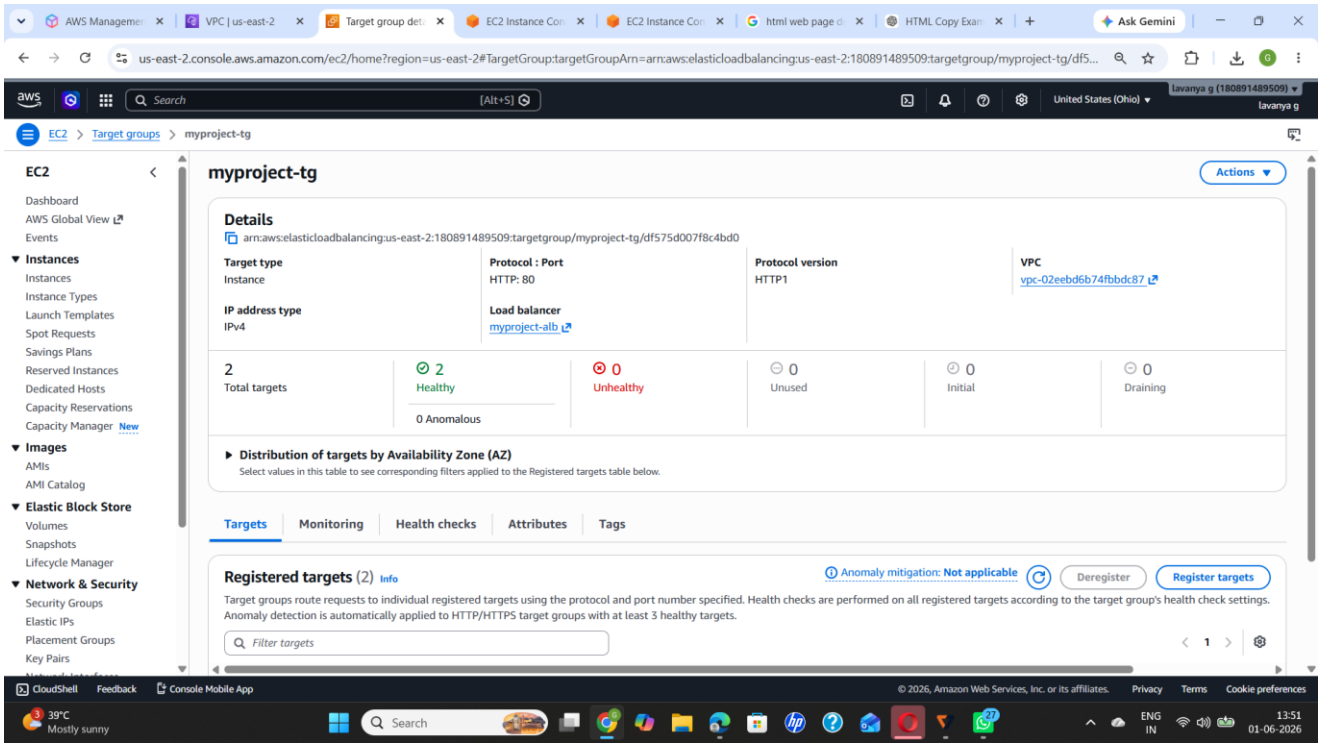
Now we need to create target group

First create a target group

1.give name (myproject-tg)

2. select 2 public instances, then click on create target group.





Now create application load balancer.

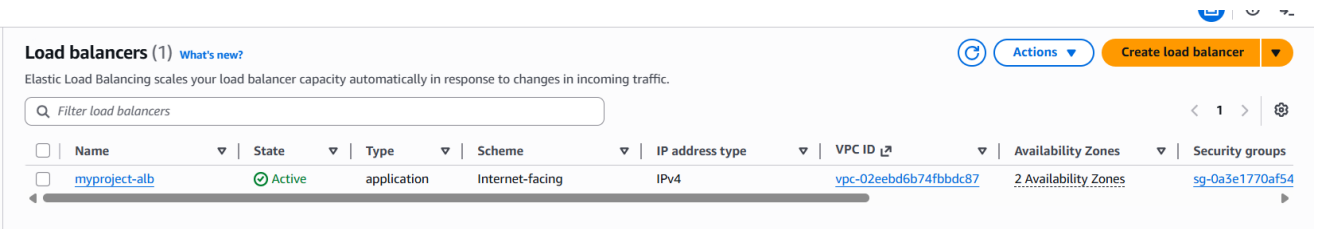
1.name (myproject-alb)

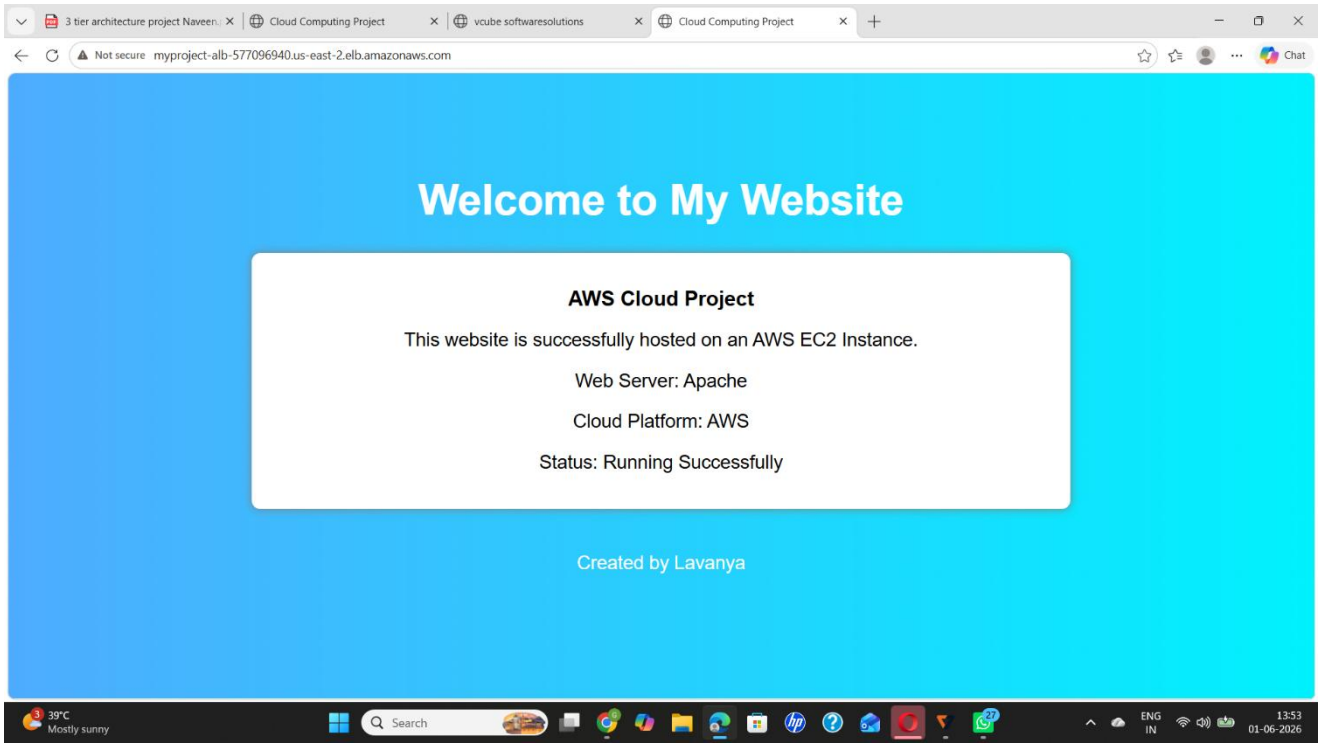
2.select created vpc, then availability zones.

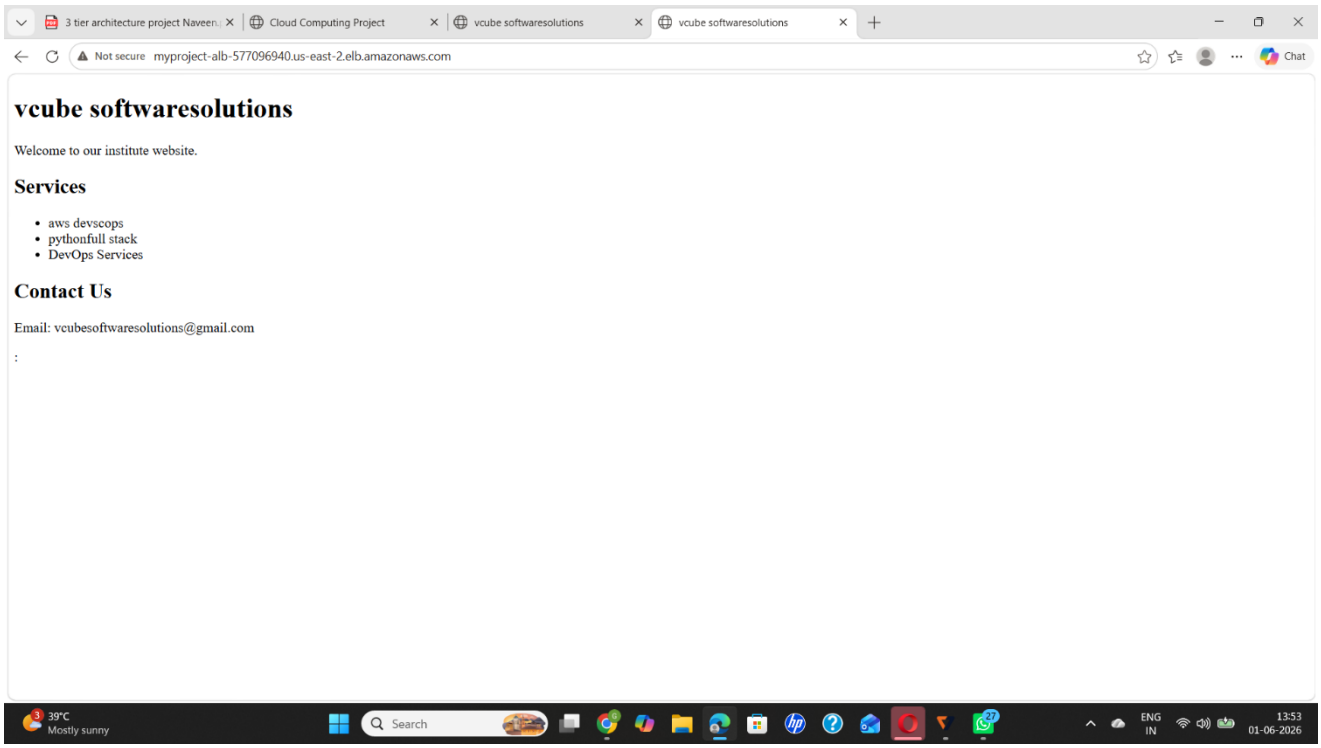
3.select security group (projectsg).

4.select target groups (myproject-tg).

5.click on create load balancer. After when it is available copy the DNS name URL and paste it on the new tab it will show the output which u have entered in webservers. Do refresh it will switch between two servers because application load balancer has round robin algorithm.



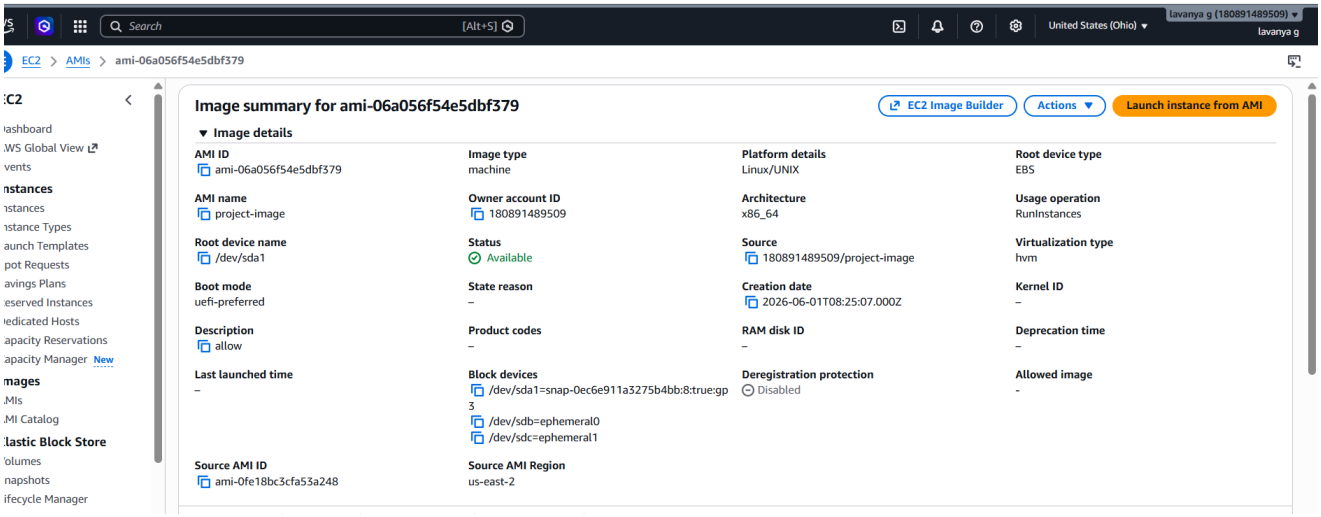




Next we need to create AMI (amazon machine image).

Go to public instance 1 click on actions select image and templates in that click on create image.

1.Enter name (project-image)



2.click on create image now go to AMI's check whether created image is available or not.

Now create Auto scaling group to create ASG we need to have launch template. First create launch template then go and create Auto Scaling Group.

Launch template:

1.Name.

2.select AMI.

3.select keypair, instance type, security group.

4.click on create launch template.

Auto Scaling Group:

1.Name.

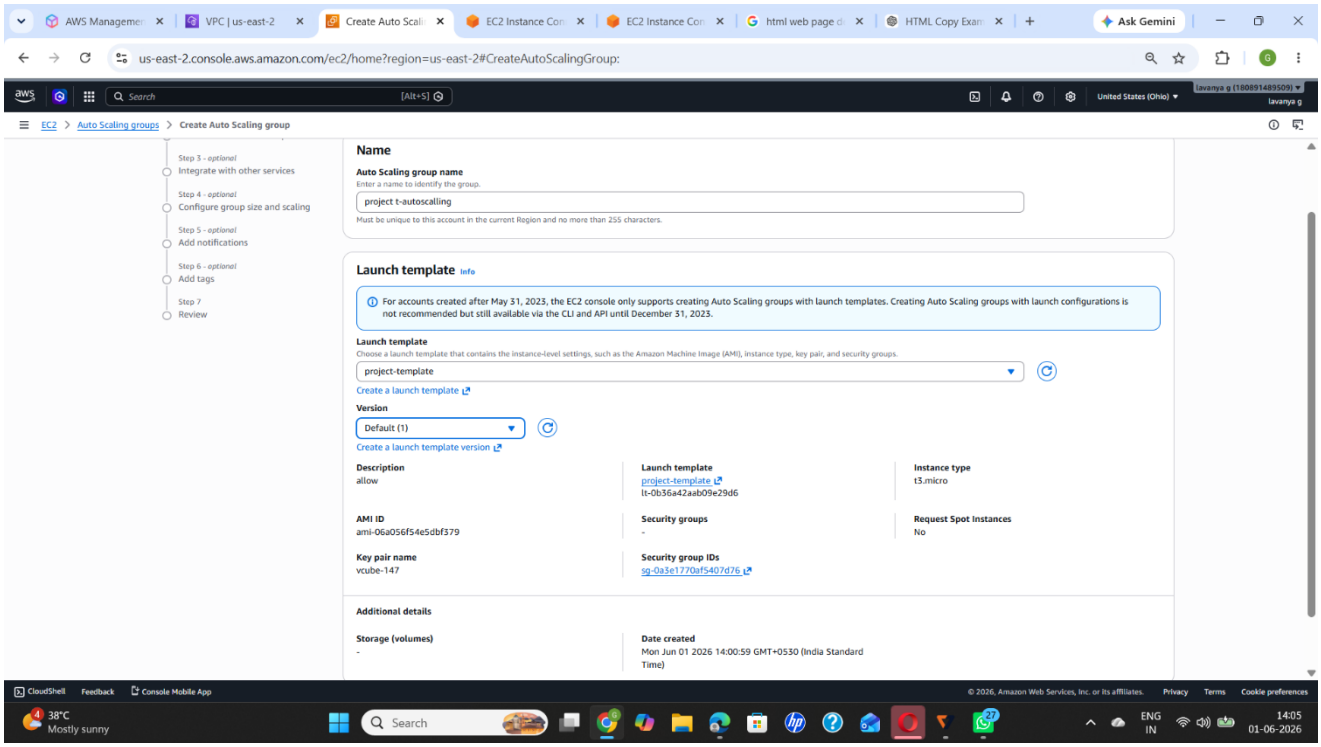
2.select launch template (select created launch template).

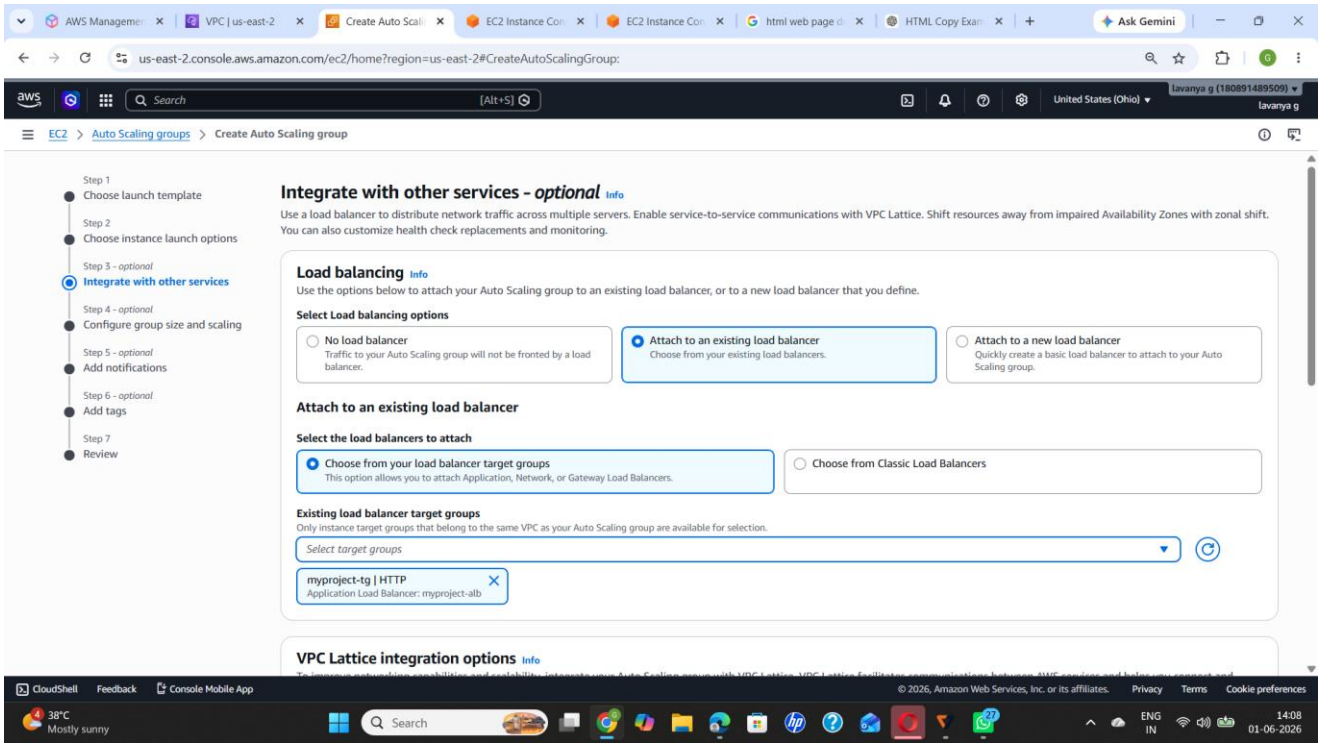
3.select vpc, availability zones, subnets

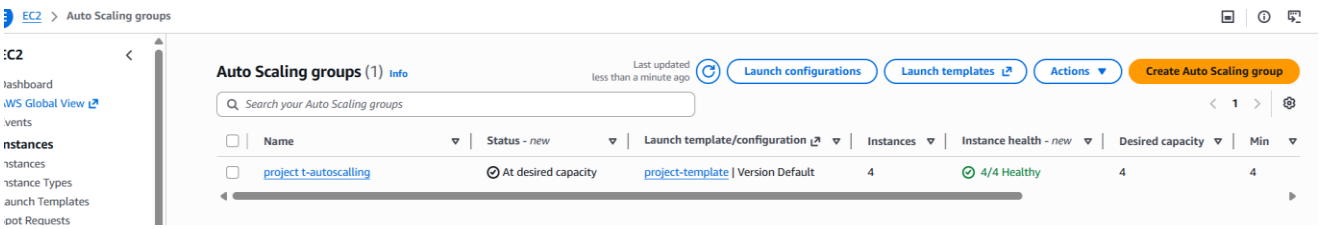
4.select existing load balancer, target groups

5. select desired capacity as per your requirement I have taken 4, In target tracking scaling policy set target values (50) and instance warmup (40) as per your requirement.

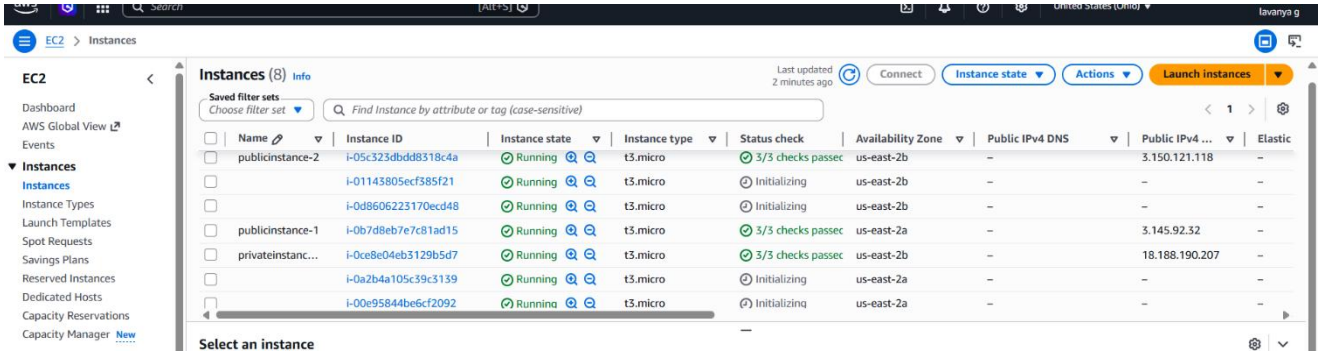
6.Click on create auto scaling group.







Now go to ec2 instances there we have another 4 ec2 instances created by auto scaling group.



Step 3:

We need to create RDS database. Before creating it first create subnet groups.

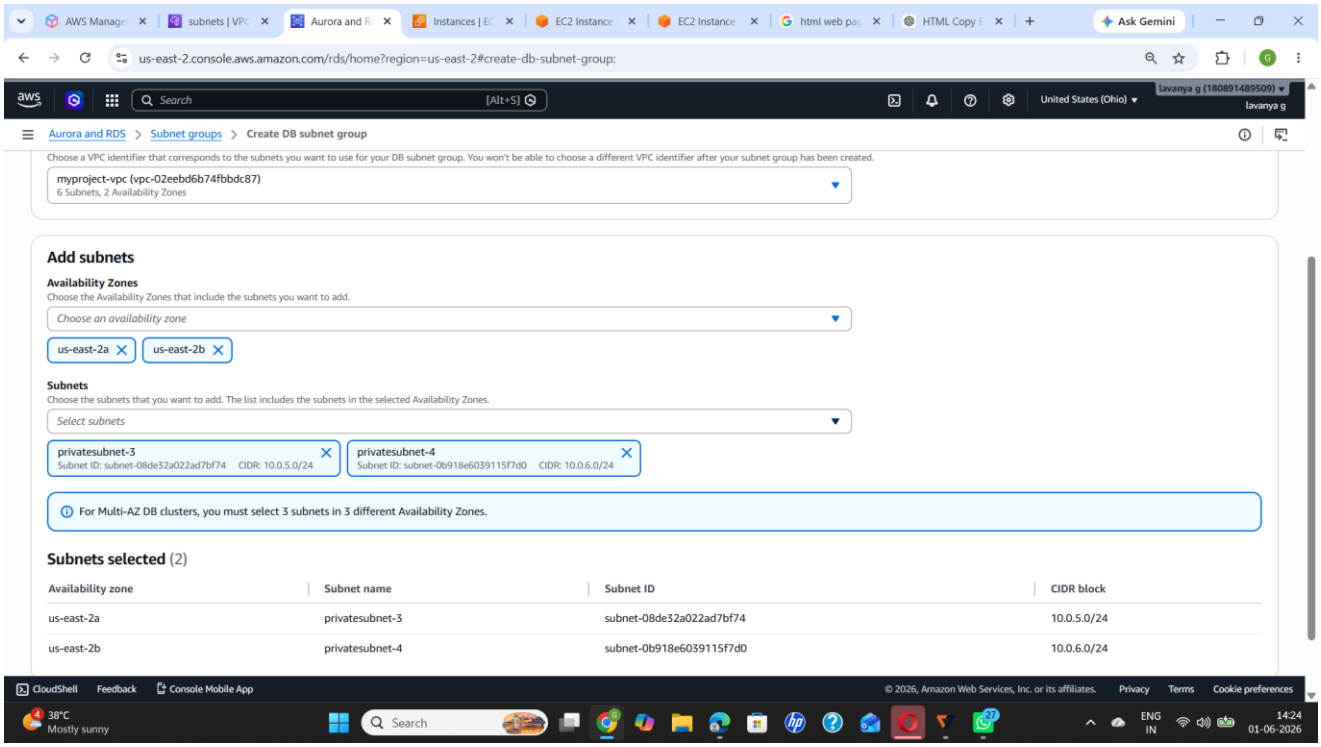
Subnet groups:

1.name (my-private-subnetgroup), description (allow)

2. select vpc (created my-project-vpc)

3. select availability zones, subnets (private subnets 3,4).

4.click on create subnet groups.

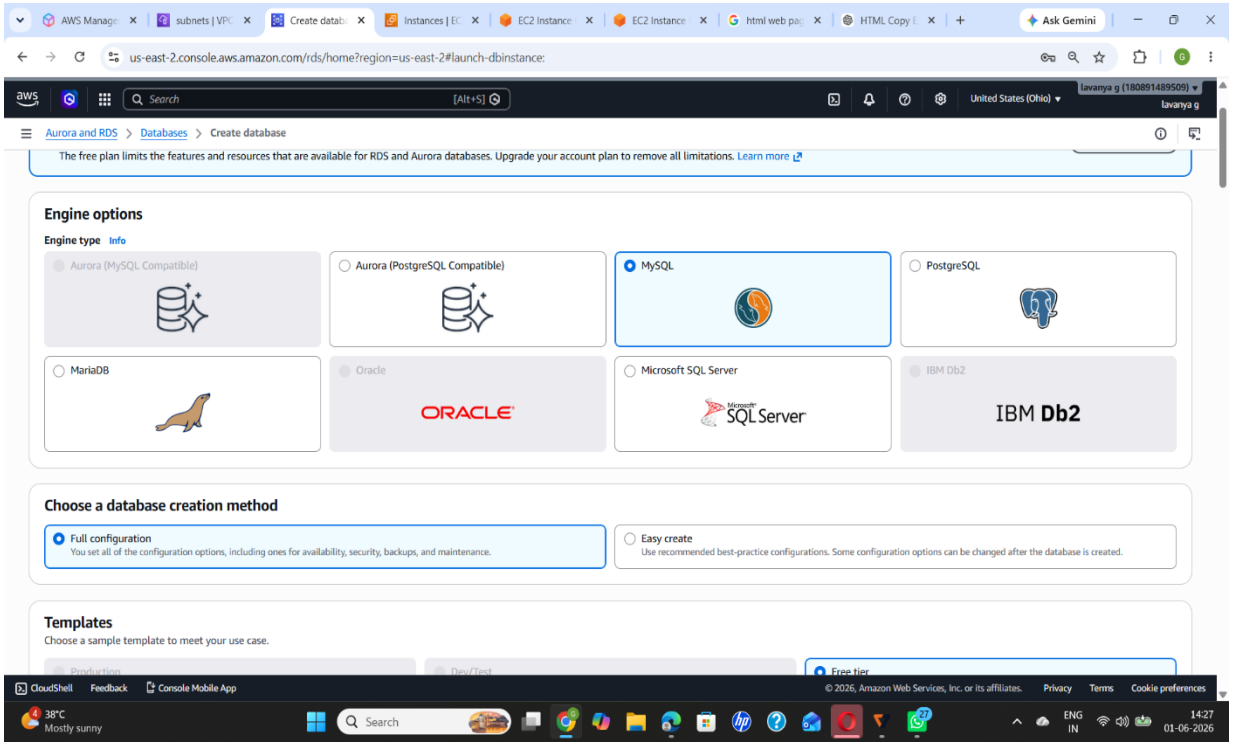


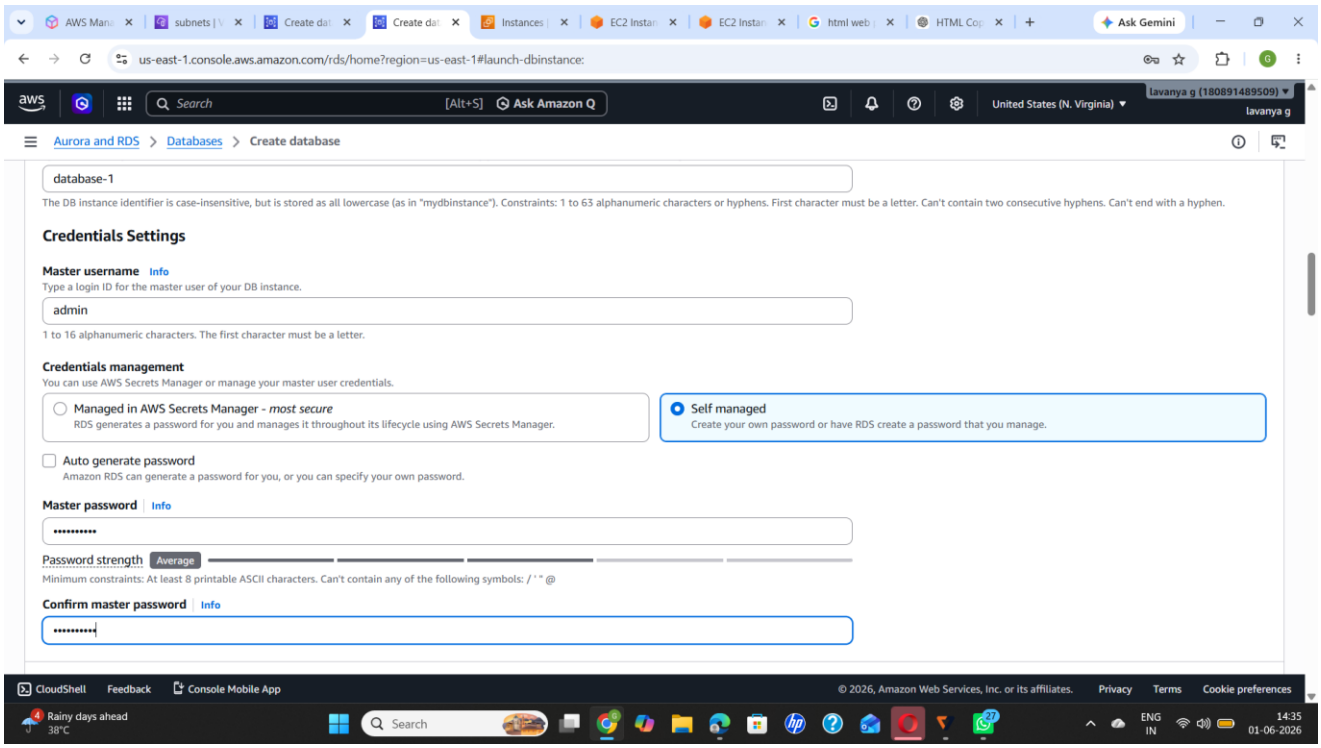
RDS:

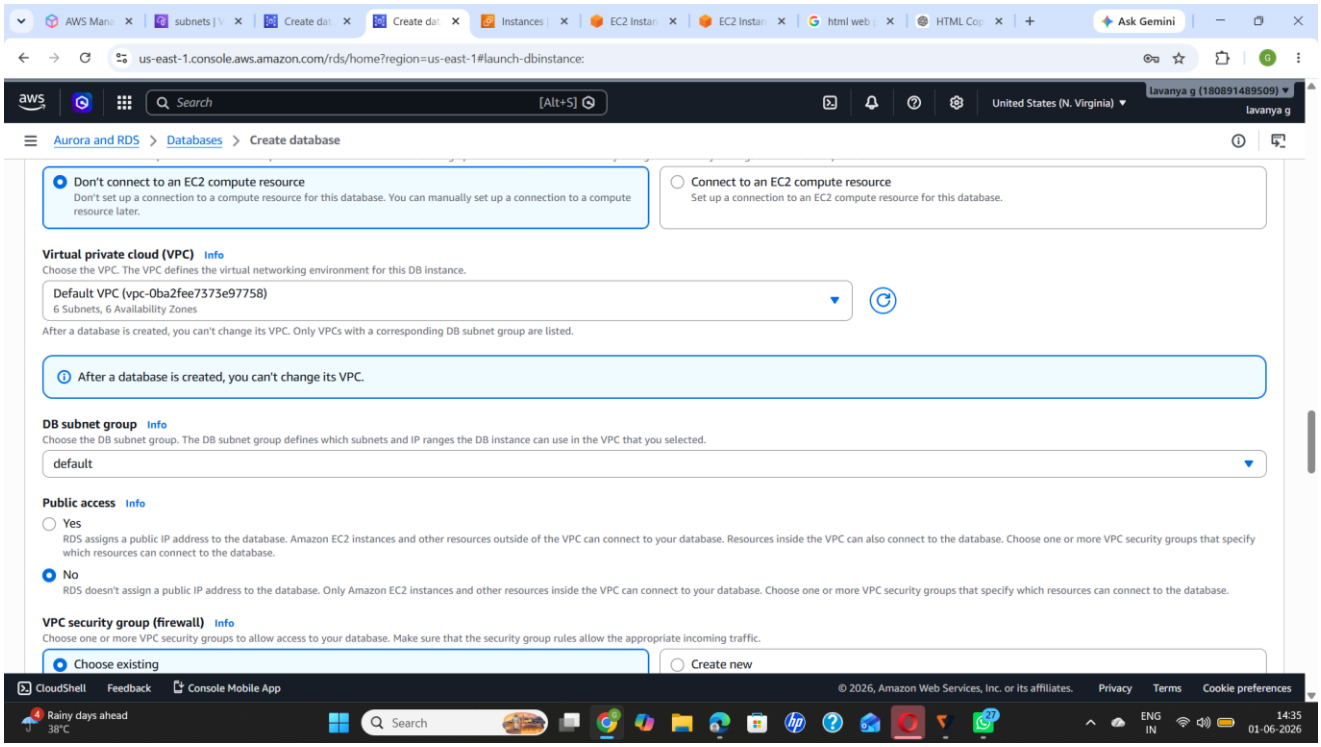
1. Click on create database, select standard create, select engine type as MySQL.
2. Select templates as production and select multi-AZ DB cluster.
3. Select on self-managed, give password and confirm the password.
4. Select memory optimized class.
5. In connectivity, click on Don't connect to the EC2 compute resource and select created vpc (my-projectvpc).
6. Select subnet group (myprivate-subnetgroup) and give public access as yes.
7. Choose existing security group (projectsg).

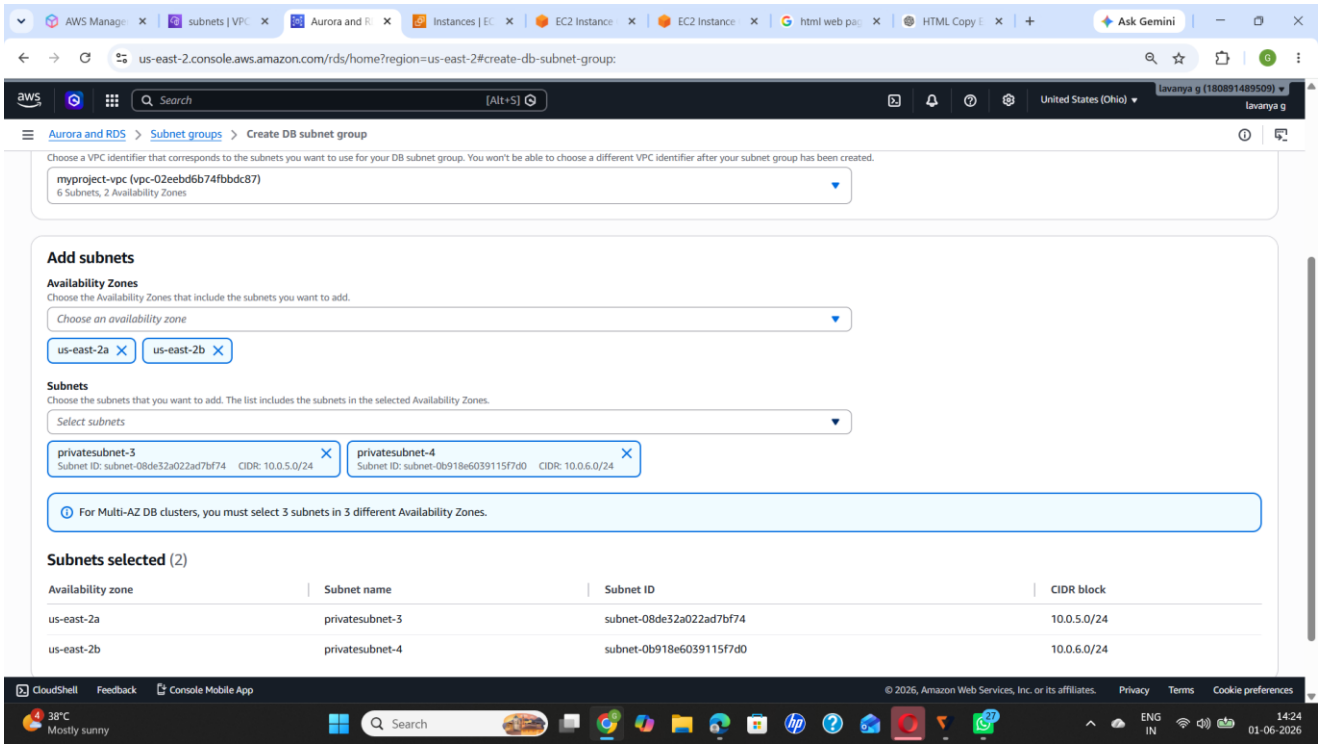
8. Go to VPC dashboard, click on VPC, click on actions, go to edit VPC settings and click on the enable DNS hostnames.

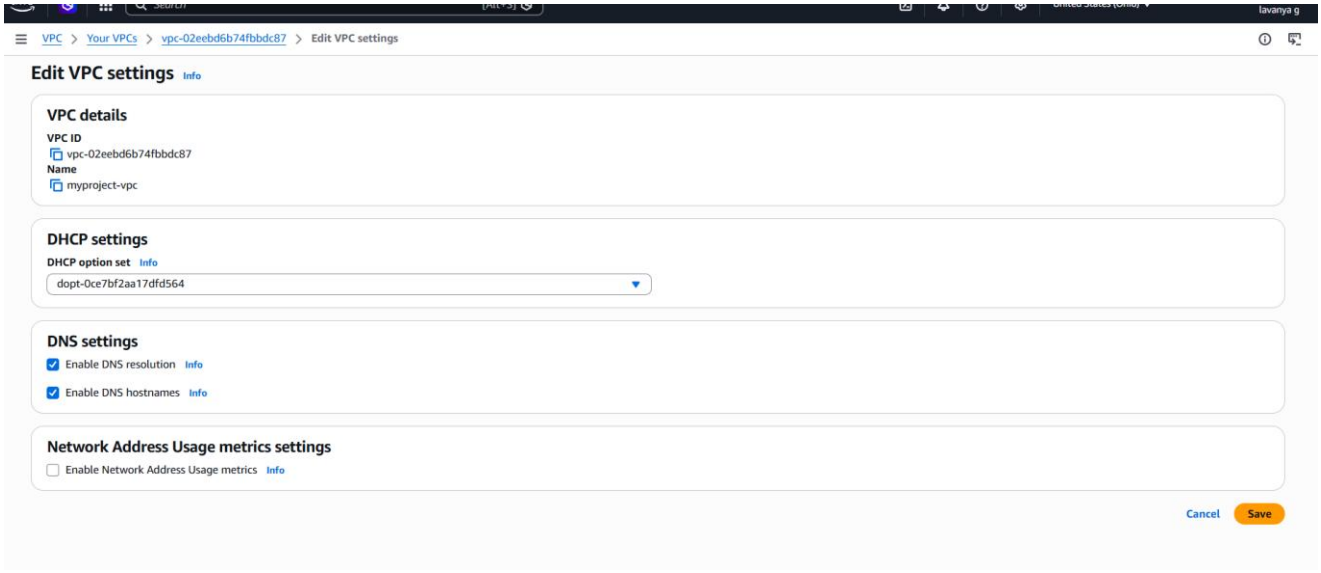
9. Create the database.






us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#launch-dbinstance:
Search
[Alt+S]
United States (N. Virginia)
lavanya g (180891489509)
lavanya g
Aurora and RDS > Databases > Create database
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
Default VPC (vpc-0ba2fee7373e97758)
6 Subnets, 6 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
default
Public access Info
Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Create new
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Rainy days ahead
38°C
ENG
IN
14:35
01-06-2026


us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#create-db-subnet-group:
Search
[Alt+S]
United States (Ohio)
lavanya g (180891489509)
lavanya g
Aurora and RDS > Subnet groups > Create DB subnet group
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
myproject-vpc (vpc-02eebd6b74fbbdc87)
6 Subnets, 2 Availability Zones
Add subnets
Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
us-east-2a
us-east-2b
Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
privatesubnet-3
Subnet ID: subnet-08de32a022ad7bf74 CIDR: 10.0.5.0/24
privatesubnet-4
Subnet ID: subnet-0b918e6039115f7d0 CIDR: 10.0.6.0/24
For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.
Subnets selected (2)
Availability zone
Subnet name
Subnet ID
CIDR block
us-east-2a
privatesubnet-3
subnet-08de32a022ad7bf74
10.0.5.0/24
us-east-2b
privatesubnet-4
subnet-0b918e6039115f7d0
10.0.6.0/24
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
38°C
Mostly sunny
ENG
IN
14:24
01-06-2026



After creating RDS login into public instance so that we are able to connect to the private instance.

Steps:

1. vi keypair.pem
2. open pem file copy the data and paste it on vi key pair.pem
3. chmod 400 keypair.pem

4.now go to private instance and copy SSH url and paste it and Enter.

5.after logging we need to install mysql on servers Steps:

1.Apt install mysql-server

2.Systemctl start mysql.service

After installing we need to login into mysql

Mysql -h endpoint url -u admin -p

Then enter password

Now you will be logged into mysql

1.show databases;

2.create database lavanya;

3.use lavanya;

4. CREATE TABLE Persons (
  PersonID int PRIMARY KEY,
  LastName varchar(255) NOT NULL,
  FirstName varchar(255),
  Address varchar(255),
  City varchar(255)
);

5. we can insert data into that table using this command

6. INSERT INTO Persons (PersonID, LastName, FirstName, Address, City) VALUES (1, 'gorle', 'lavanya', ‘Kphb’, ‘Hyderabad’);

7. show tables; (to show tables in that DB)

8.Select * from Persons;

Now exit from private instance, we need to login into mysql from public instance .same process install mysql then login ,enter password.

1.show databases;

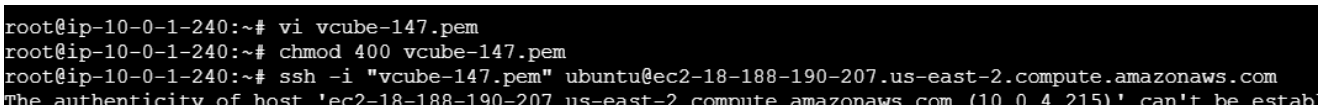
It will show all databases including which we have created in private instances.

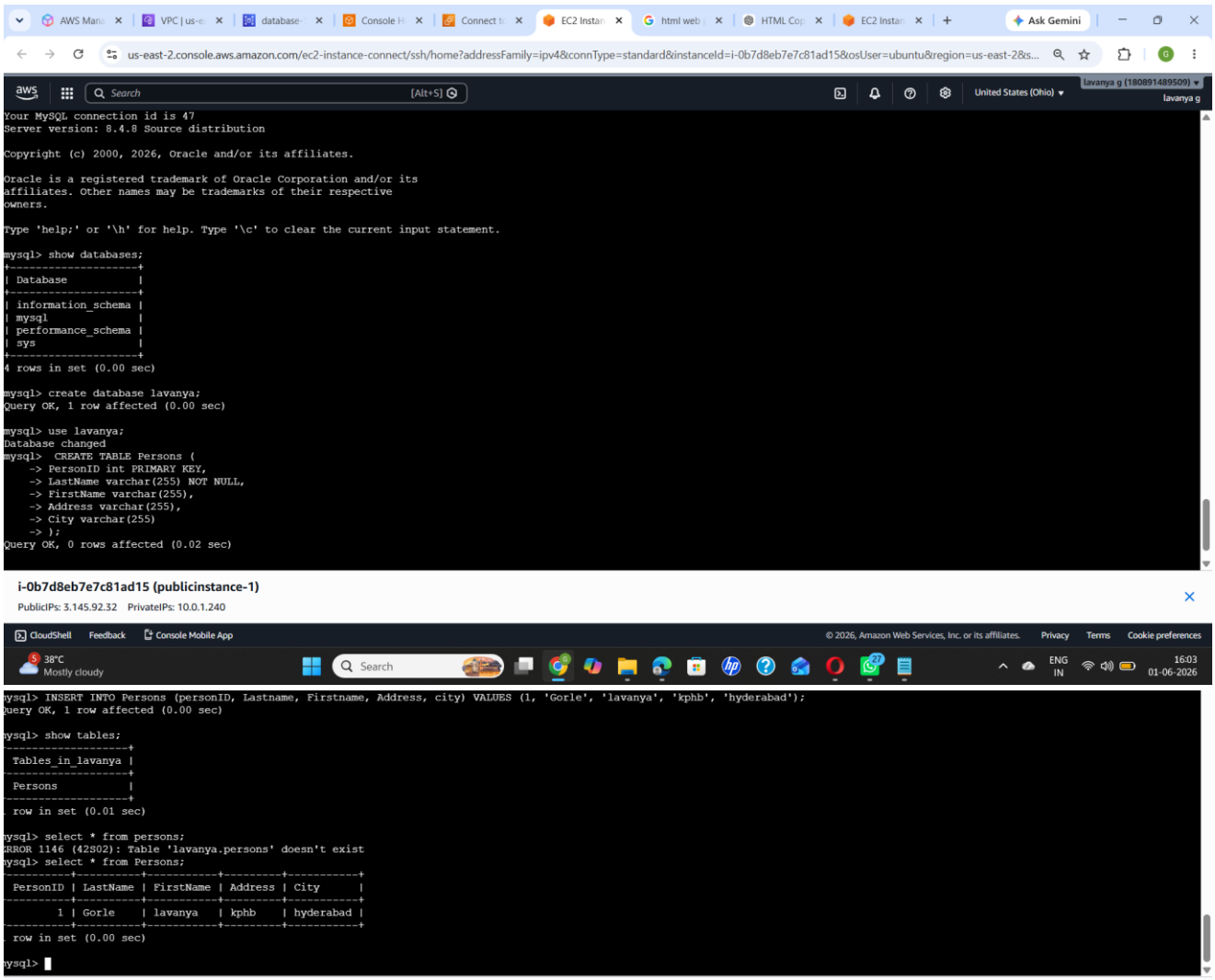
2.use lavanya;

3.show tables;

4.select * from Persons;

It displays the information which we inserted previously.





```
mysql> exit
Bye
root@ip-10-0-4-215:~# mysql -h database-1.c7scoaokq0c5.us-east-2.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 59
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| lavanya            |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.01 sec)

mysql>
```

```
mysql> use lavanya;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-------------------+
| Tables_in_lavanya |
+-------------------+
| Persons           |
+-------------------+
1 row in set (0.01 sec)

mysql> slect * from Persons;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for t
mysql> select * from Persons;
+----------+----------+-----------+---------+-----------+
| PersonID | LastName | FirstName | Address | City      |
+----------+----------+-----------+---------+-----------+
|        1 | Gorle    | lavanya   | kphb    | hyderabad |
+----------+----------+-----------+---------+-----------+
1 row in set (0.01 sec)

mysql>
```

We have created target groups, load balancers ,auto scaling groups in public instances. Now we have to do the same process for private instances in order to create target groups, load balancers, auto scaling groups ..